

Joby

Overset Grid Usage at Joby Aviation

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jobyaviation.com

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Imagine

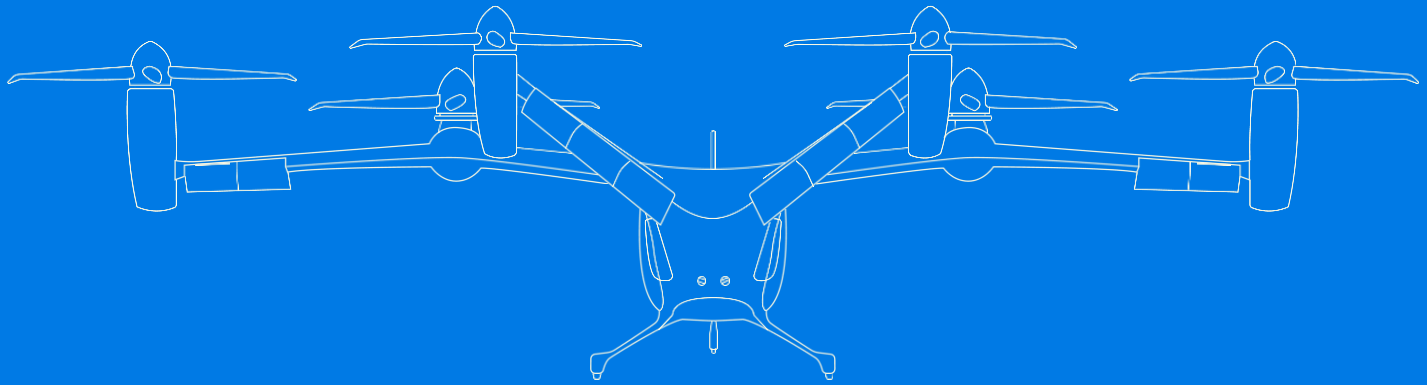
Turning your commute into a serene, stress-free journey through the clouds.



Joby Air Taxi Specifications

What if you could turn traffic into thin air? It's possible with our smooth and fast clean energy aircraft. Think of it as your childhood dream, all grown up.

Back view

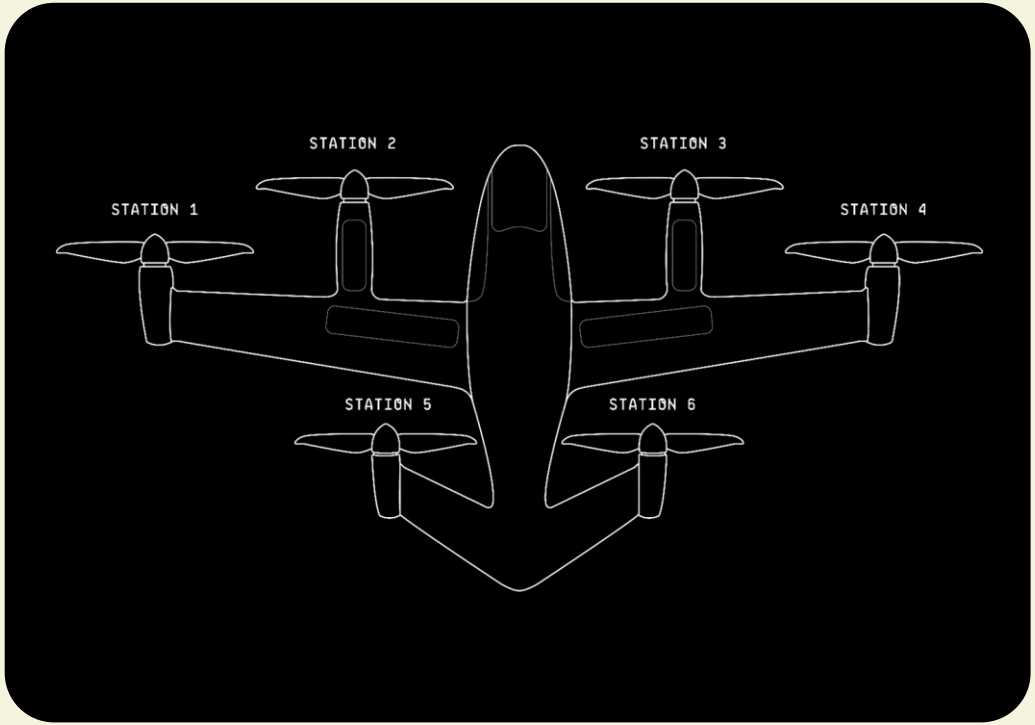


01	Passengers	Holds 5 (4 passengers + Pilot)
02	Speed	Up to 200 mph (174 knots)
03	Sound	As quiet the wind in the trees (45 dBA) flyover at 1500 ft altitude
04	Electric	0 operating emissions

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Redundancy built-in

We designed our aircraft's powertrain, flight controls, and electronics to be fully redundant, with six dual-wound propulsion motors, four battery backs, a triple redundant flight control computer and dual critical actuation systems.



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Service & Infrastructure Partners

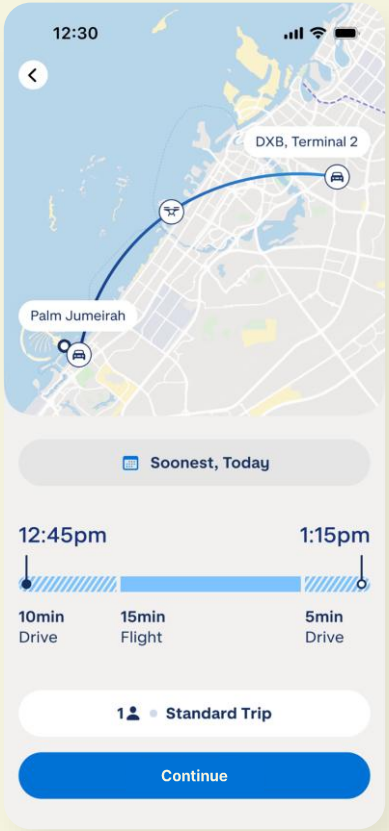
We're partnering with global leaders in ground transportation, airlines, and key infrastructure providers to seamlessly integrate our air taxi service into the future of door-to-door travel while building the physical backbone needed for convenient everyday flight.

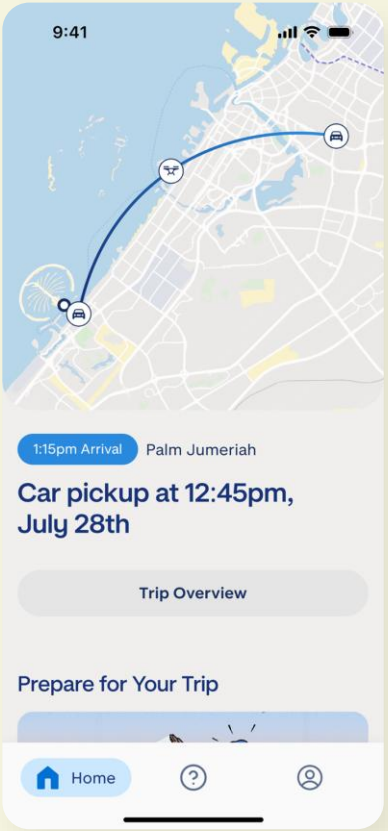


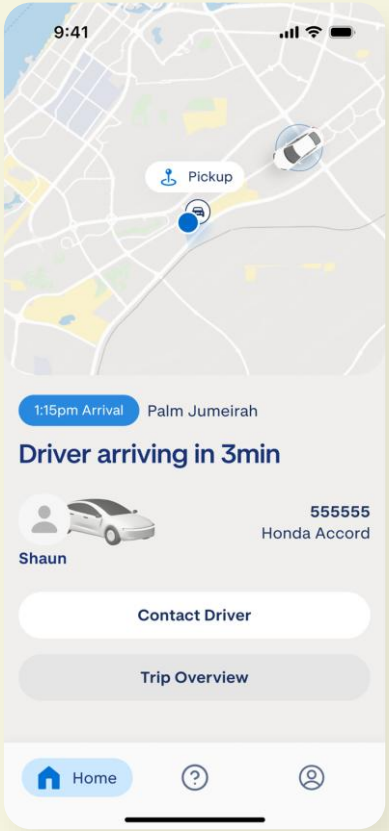
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The Joby App

A few taps on our app and we'll choreograph your entire commute, including coordinating an Uber to and from our vertiport. No hassle or headaches, just smooth sailing from A to B.



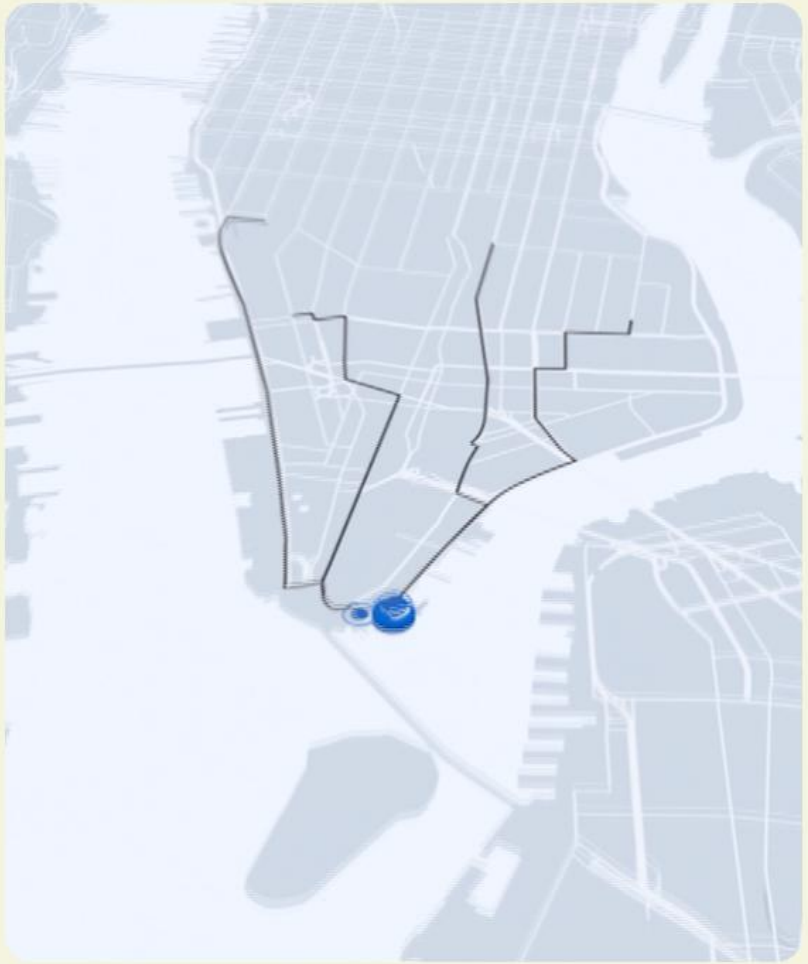




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Faster Mobility

Riders will be able to reach their destination up to 10x faster at busy times.

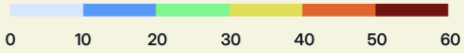


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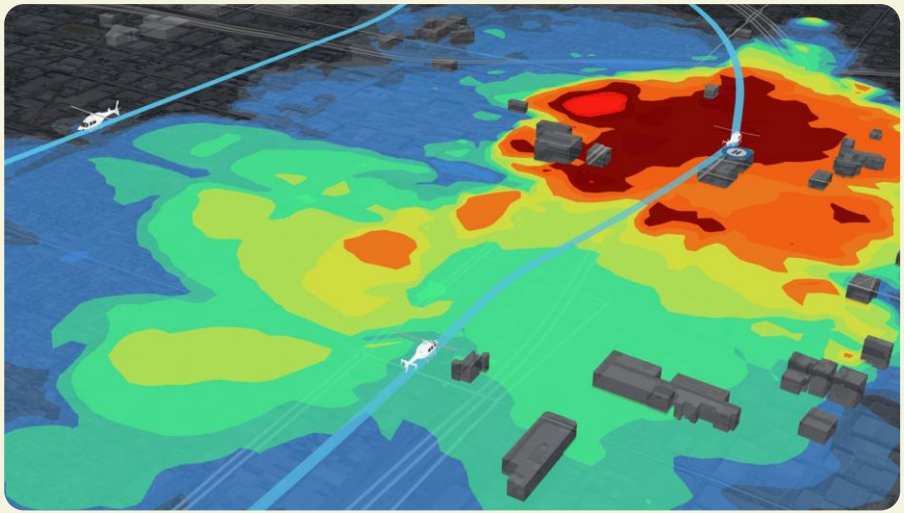
Revolutionary Acoustic Design

Low noise design for reduced acoustic impact in urban environments

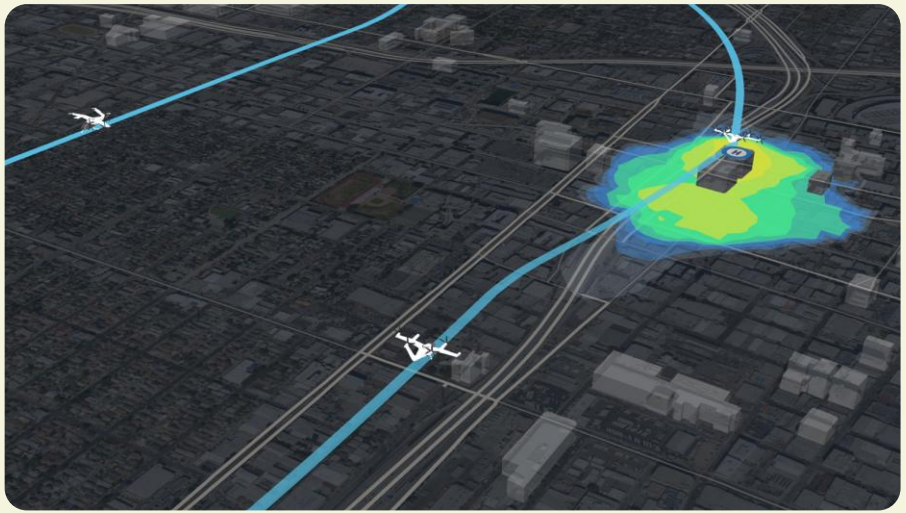
Time above ambient noise (sec)



Helicopter acoustic footprint



Joby air taxi acoustic footprint



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Tilting Propellers

Six propeller stations rotate (3 in each direction)

Each of the three pairs have different nacelles

Five blades on each propeller

Ideal candidate for overset meshing



(a) Thrustborne flight



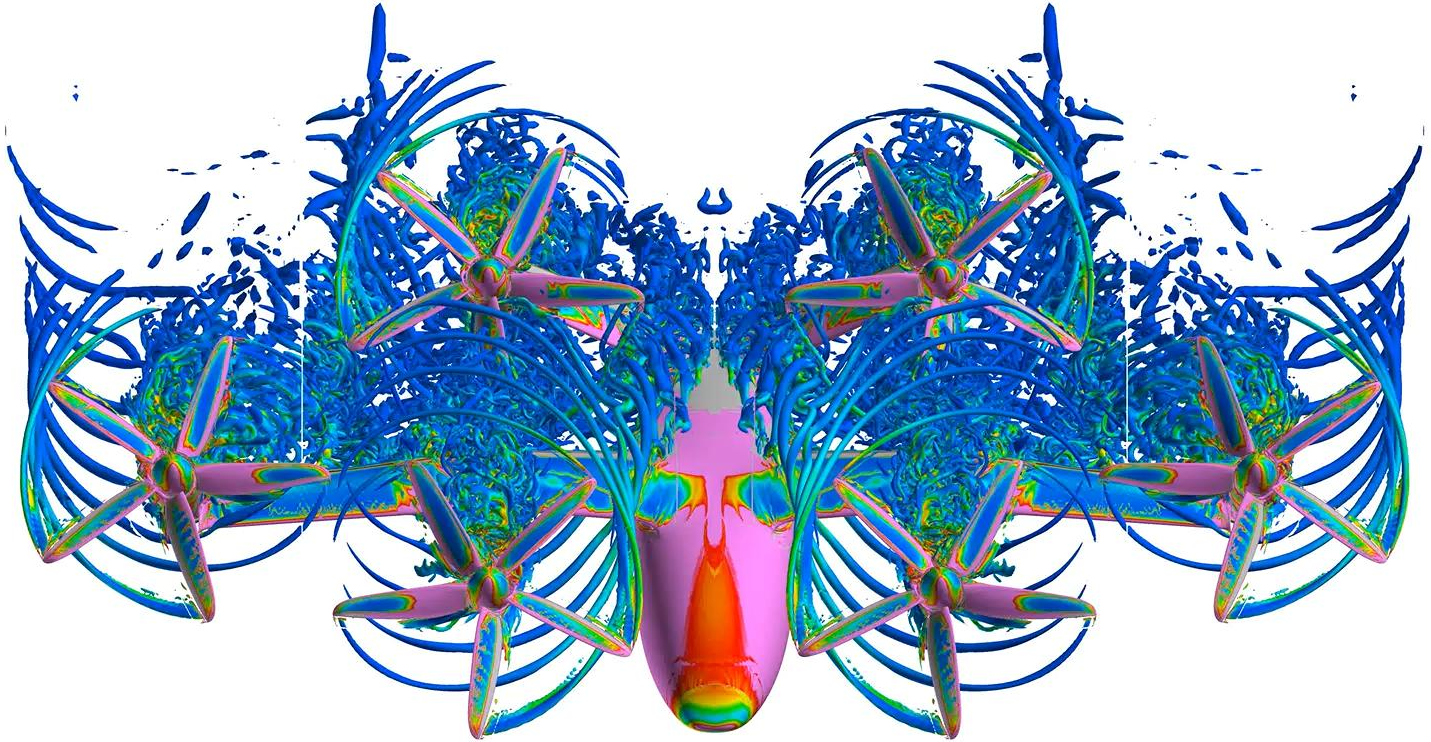
(b) Semi-thrustborne flight



(c) Wingborne flight

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Flow Field in Transition

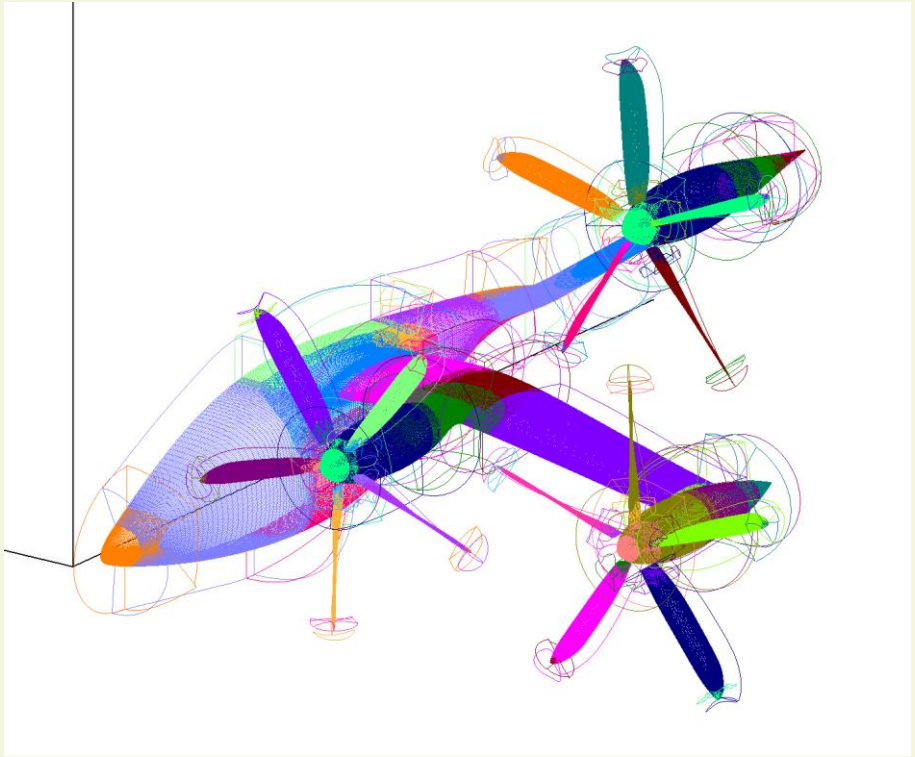


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2017 Full Scale Prototype

Model only created for cruise flight
Manually generated
Grid Connectivity with Xrays
120 Million grid points run on local cluster



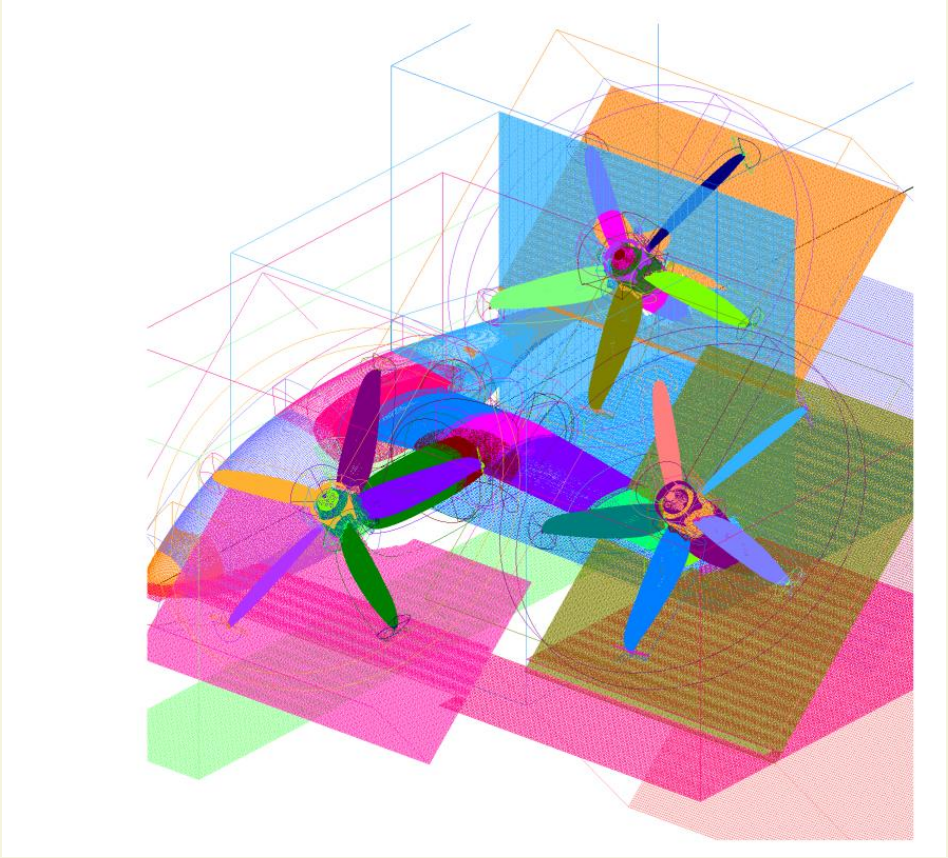


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2019 Pre-Production Prototype

Scripted grid generation
Models with and without flap
Tilting geometry
Compatibility with rotor disk model
Hybrid grid connectivity with Pegasus
220 Million Pts run on AWS GovCloud



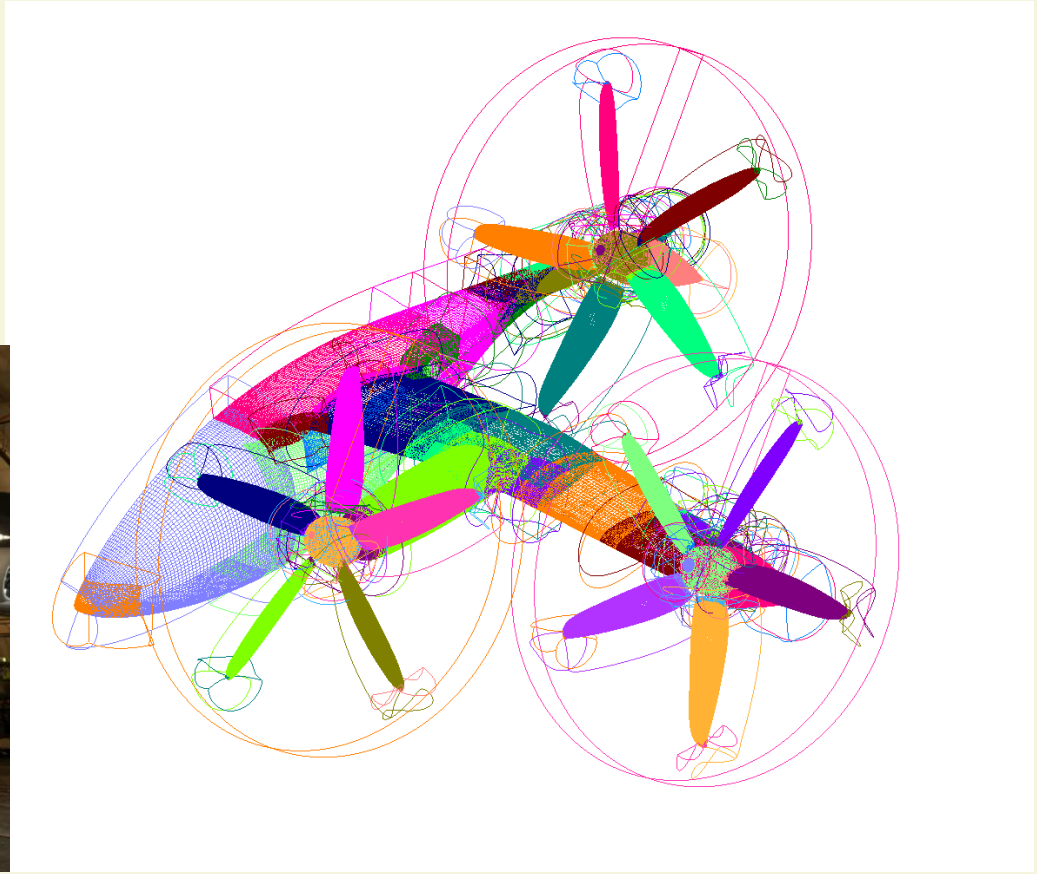


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2023 Production Prototype

- Scripted grid generation
- Moving control surfaces
- Tilting geometry
- Hybrid grid connectivity with Pegasus
- Individual integrated and distributed component forces, moments



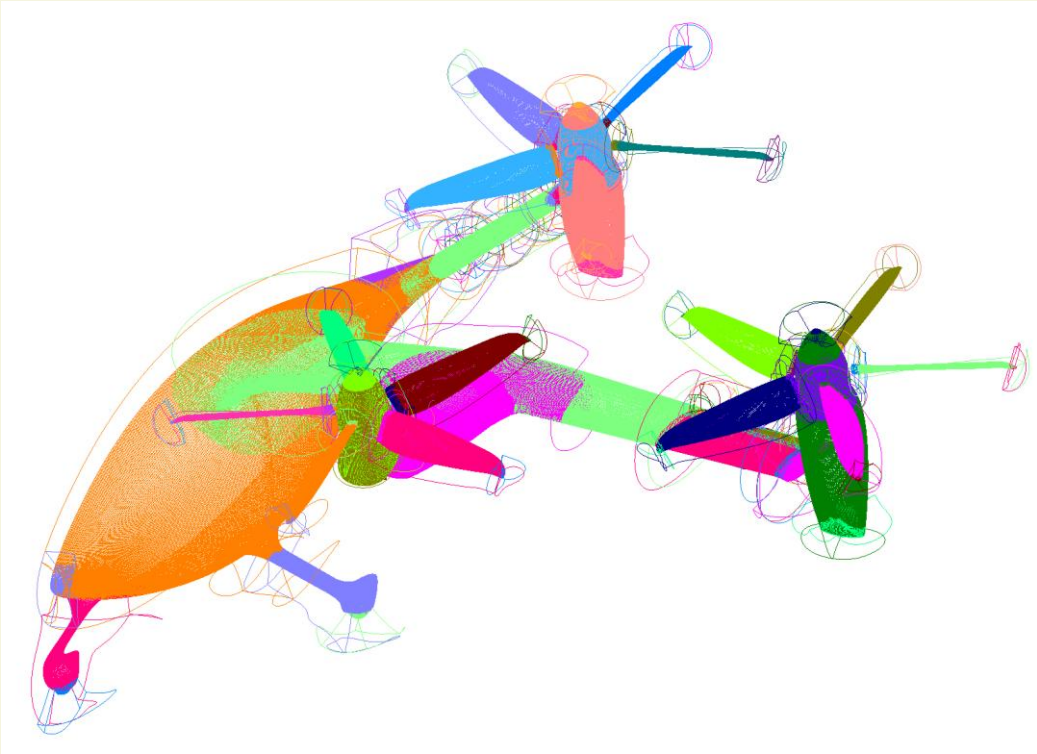


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2026 FAA Conforming

- Scripted grid generation
- Moving control surfaces
- Landing gear
- Tilting geometry
- Grid connectivity with xrays
- 260 Million Pts run on Local Cluster





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2026 FAA Conforming - Helios

- ROAM and Blade Resolved
- Nacelle, control surface, and blade motion
- Forces computed on individual components
- Rapidus near body and all running on GPU



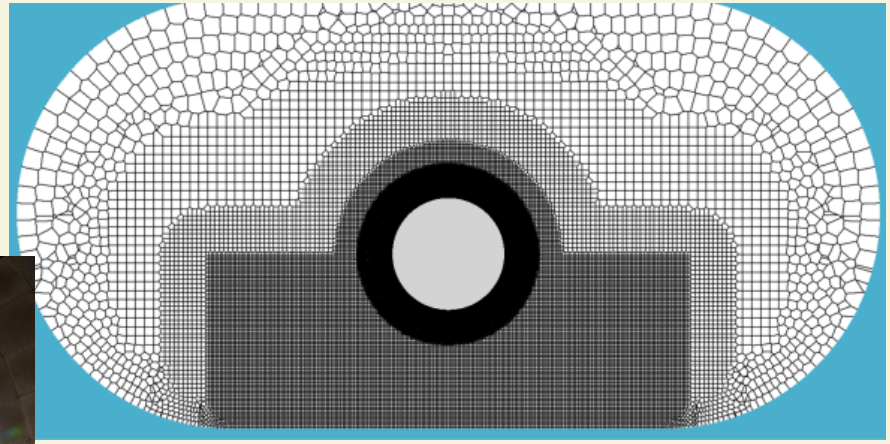
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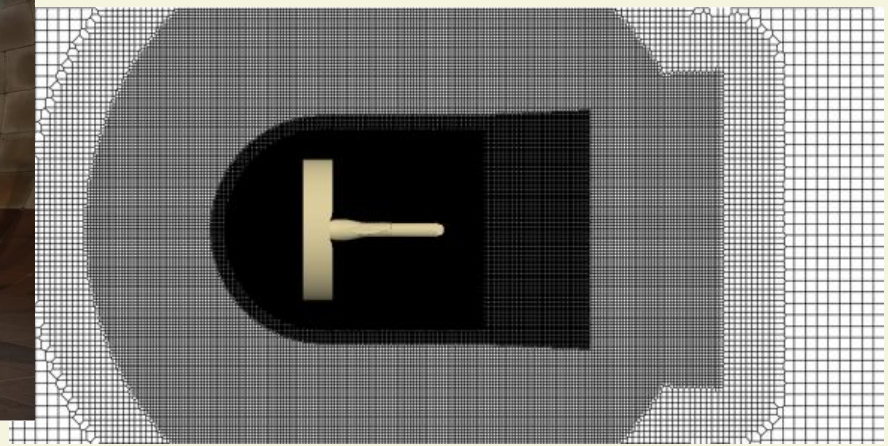
Isolated Propeller in NFAC

Analysis using Overflow, CharLES, Flow360, and Helios of the Joby propeller installed at NFAC

Some analysis used different fidelity of the nacelle, sting, and wind tunnel walls





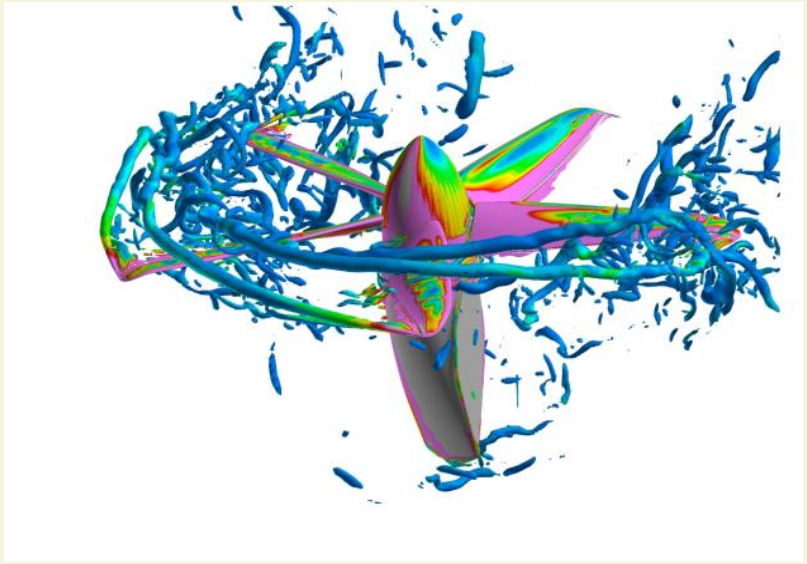


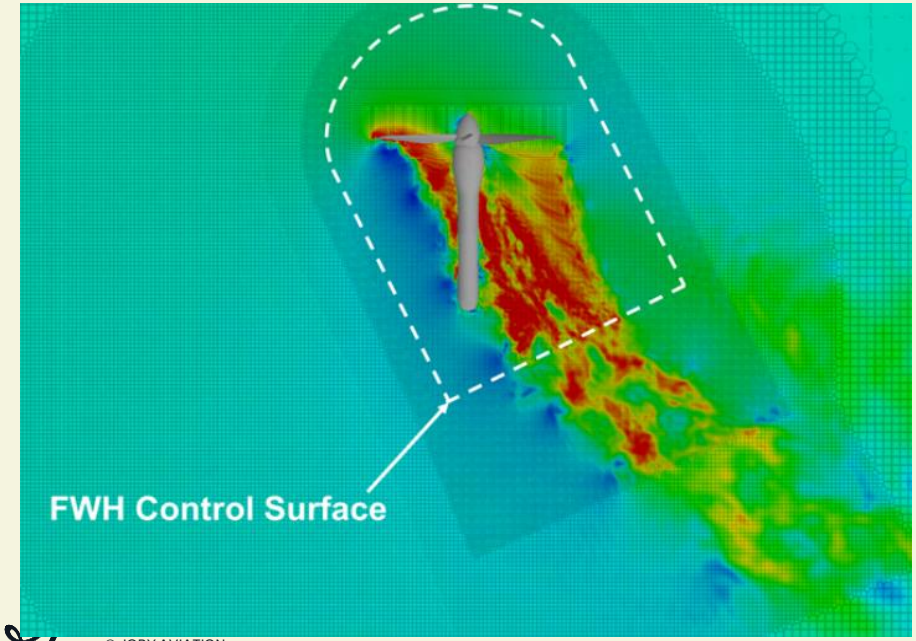
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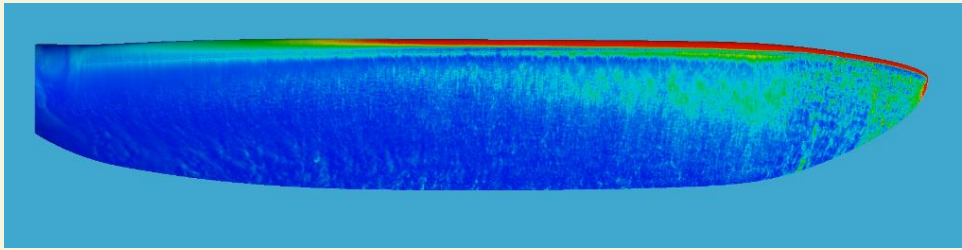
Isolated Propeller Acoustic Validation

Overflow with DDES for high fidelity wake

LES (CharLES) shows promise for predicting first principal boundary layer noise sources







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Areas of Possible Software Improvements

Continued development and optimizations for GPUs
- We have seen a 5x improvement hardware $/run using used V100 GPUs over CPUs

Transition modeling for GPU version of Overflow and Helios

EROTOR namelist capabilities in GPU version of Overflow

ROAM-like actuator line model capability in standard Overflow

Further enhancements for automatic structured meshing (EPOGS)

Multi-block structured grid in Rapidus

Half aircraft modeling in Rapidus

V100 support for LAVA

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